

Project Title: AI-Assisted Threat Detection

Dashboard_____

Project Statement:

This project aims to develop an AI-powered cybersecurity monitoring and threat detection platform that assists security analysts in identifying, prioritizing, and responding to potential cyber threats through intelligent analytics and real-time security monitoring. By leveraging machine learning algorithms, threat intelligence feeds, anomaly detection techniques, and security event analytics, the platform provides actionable insights into organizational cybersecurity risks.

The system continuously analyzes security events, attack indicators, vulnerability information, and threat intelligence data to identify suspicious activities, emerging attack patterns, and high-risk security incidents. Through AI-assisted risk scoring and threat prioritization mechanisms, cybersecurity teams can focus on the most critical threats and improve incident response effectiveness.

An interactive dashboard serves as the central monitoring interface, providing visibility into threat severity levels, attack trends, security posture metrics, and operational risk indicators._____

Outcomes:

- AI-assisted identification and prioritization of cybersecurity threats.
- Automated threat scoring and risk assessment capabilities.
- Detection of abnormal security activities and attack patterns.
- Real-time visibility into organizational security posture.
- Improved threat response and cybersecurity decision-making through actionable intelligence.

Key Technologies:

- Python: Pandas, NumPy, Scikit-learn
- Visualization: Plotly/Dash, Power BI, Tableau Concepts
- Machine Learning: Anomaly Detection, Classification Models
- Security Frameworks: MITRE ATT&CK, CVE Intelligence, Threat Intelligence Feeds

Modules to be Implemented:

Module 1: Security Data Aggregation & Threat Intelligence Layer

- Collect and process security logs, vulnerability feeds, threat intelligence sources, and incident records.
- Normalize and enrich security events for downstream analysis.
- Map security indicators to cybersecurity frameworks such as MITRE ATT&CK.

Module 2: AI-Based Threat Detection & Anomaly Analysis Engine

- Identify abnormal security behavior and suspicious activity patterns using machine learning techniques.
- Detect unusual login activity, attack indicators, threat signatures, and risk anomalies.
- Generate threat confidence scores and incident severity classifications.

Module 3: Risk Prioritization & Security Intelligence Module

- Correlate threat events from multiple sources to identify high-risk attack scenarios.

- Calculate risk scores based on threat severity, asset criticality, and vulnerability exposure.

- Generate recommendations for incident prioritization and response actions.

Module 4: Threat Monitoring Dashboard & Security Reporting Platform

- Provide real-time monitoring dashboards for threat visibility and operational awareness.
- Display attack trends, threat distributions, risk scores, and security intelligence insights.
- Support filtering, drill-down analysis, and executive cybersecurity reporting.

Project Milestones:

- Milestone 1 - Weeks 1–2: Security Data Aggregation & Threat Intelligence Layer

Objective: Develop the data ingestion and threat intelligence integration framework.

Tasks:

- o Collect and preprocess cybersecurity datasets from logs, threat feeds, and vulnerability repositories.
- o Standardize security event formats and enrichment processes.
- o Establish threat intelligence mapping and classification mechanisms.

- Milestone 2- Weeks 3–4: AI-Based Threat Detection & Anomaly Analysis Engine

Objective: Develop intelligent threat detection and anomaly identification capabilities.

Tasks:

- o Implement machine learning models for anomaly detection.
- o Identify suspicious activity patterns and attack indicators.
- o Generate automated threat classifications and confidence scores.

- Milestone 3 - Weeks 5–6: Risk Prioritization & Security Intelligence Module

Objective: Develop threat correlation and cybersecurity risk assessment functionality.

Tasks:

- o Implement risk scoring algorithms and event correlation mechanisms.
- o Generate threat prioritization insights and response recommendations.
- o Validate threat detection performance using simulated attack datasets.

- Milestone 4 – Week 7-8: Threat Monitoring Dashboard & Security Reporting Platform

Objective: Develop visualization dashboards and operational cybersecurity reporting capabilities.

Tasks:

- o Build interactive threat monitoring dashboards.
- o Integrate threat intelligence, risk scoring, and security analytics modules.
- o Deploy and validate the complete cybersecurity monitoring platform.

Evaluation Criteria:

Week 1-2 (Module 1): Security datasets successfully aggregated, normalized, and enriched for analysis.

Week 3-4 (Module 2): Threat detection engine accurately identifies anomalies and suspicious security activities.

Week 5-6 (Module 3): Risk scoring and threat prioritization capabilities generate meaningful cybersecurity intelligence.

Week 7-8 (Module 4): Fully integrated AI-Assisted Threat Detection Dashboard deployed with real-time monitoring, threat analytics, and reporting functionalities. _____